

Section A: Conceptual Understanding

1. What is a CSS Pseudo Class?

A pseudo-class styles an element in a special state (hover, focus, a

Difference:

- * `.class`` → styles elements all the time.
- * `:pseudo-class`` → styles elements only in a specific state.

2. Use of `:hover`, `:active`, and `:focus`

- * `**:hover**` → Applies style when the mouse is over an element.
- * `**:active**` → Applies style when an element is clicked.
- * `**:focus**` → Applies style when an input field is selected.

3. Purpose of `:nth-child()`

`:nth-child()` selects elements based on their position.

Uses:

- * Styling even and odd rows.
- * Highlighting every 3rd item in a list.

4. Difference between `:link` and `:visited`

- * `**:link**` → Styles an unvisited link.
- * `**:visited**` → Styles a visited link.

5. Role of `:disabled`

`:disabled`` styles form fields that cannot be edited or used by the

Section B: Output Prediction

1.

```
li:first-child{  
    color:red;  
}
```

Output: The first list item becomes red.

2.

```
input:focus{  
    border:2px solid green;  
}
```

Output: The input box gets a green border when selected.

3.

```
button:active{  
    transform:scale(0.95);  
}
```

Output: The button becomes slightly smaller when clicked.

4.

```
li:nth-child(odd){  
    background-color:lightgray;  
}
```

Output: The 1st, 3rd, 5th, etc. list items get a light gray background. -

```
<!DOCTYPE html>
<html>
<head>
  <title>Interactive UI Dashboard</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
```

```
<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Services</a>
  <a href="#">Gallery</a>
  <a href="#">Contact</a>
</nav>
```

```
<h2>Student Form</h2>
```

```
<form>
```

```
<input type="text" placeholder="Name"><br><br>
```

```
<input type="email" placeholder="Email"><br><br>
```

```
<input type="password" placeholder="Password"><br><br>
```

```
<input type="text" placeholder="Disabled Field" disabled><br><br>
```

```
<button>Submit</button>
```

```
</form>
```

```
<h2>Student List</h2>
```

```
<ul>
<li>Item 1</li>
<li>Item 2</li>
<li>Item 3</li>
<li>Item 4</li>
<li>Item 5</li>
<li>Item 6</li>
<li>Item 7</li>
<li>Item 8</li>
<li>Item 9</li>
<li>Item 10</li>
</ul>
```

```
<h2>Profile Card</h2>
```

```
<div class="card">
```

```
<div class="image">Image</div>
```

```
<h3>Student Profile</h3>
```

```
<p>This is a sample profile card.</p>
```

```
<button>Read More</button>
```

```
</div>
```

<h2>Profile Card</h2>

<div class="card">

<div class="image">Image</div>

<h3>Student Profile</h3>

<p>This is a sample profile card.</p>

<button>Read More</button>

</div>

CSS-:

```
body{
  font-family:Arial;
  background:#f5f5f5;
  text-align:center;
}
```

```
/* Navigation */
```

```
nav{
  background:#333;
  padding:15px;
}
```

```
nav a{
  color:white;
  text-decoration:none;
  margin:15px;
  padding:8px 15px;
  display:inline-block;
  transition:.3s;
}
```

```
nav a:hover{
  background:orange;
  color:black;
}
```

```
nav a:active{
  transform:scale(.9);
}
```

```
nav a:visited{
  color:lightblue;
}
```

```
input{
  padding:10px;
  width:250px;
}
```

```
input:focus{
  border:2px solid blue;
  box-shadow:0 0 8px blue;
}
```

```
input:disabled{
    background:lightgray;
}

button{
    padding:10px 20px;
}

button:hover{
    background:green;
    color:white;
}

/* List */

ul{
    width:200px;
    margin:auto;
}

li{
    padding:8px;
}

li:first-child{
    background:red;
    color:white;
}

li:last-child{
    background:blue;
    color:white;
}

li:nth-child(even){
    background:#ddd;
}

li:nth-child(3n){
    color:green;
    font-weight:bold;
}

/* Card */

.card{
    width:250px;
    margin:auto;
    background:white;
    padding:20px;
    box-shadow:0 0 10px gray;
    transition:.3s;
}
```

```
.card:hover{
  box-shadow:0 0 20px black;
}

.image{
  width:100%;
  height:150px;
  background:#ccc;
  line-height:150px;
}

/* Shadow Boxes */

.shadow-container{
  display:flex;
  justify-content:center;
  gap:20px;
  flex-wrap:wrap;
}

.box{
  width:100px;
  height:100px;
  background:white;
  line-height:100px;
}

.bottom{
  box-shadow:0 8px 10px gray;
}

.top{
  box-shadow:0 -8px 10px gray;
}

.left{
  box-shadow:-8px 0 10px gray;
}

.blur{
  box-shadow:0 0 20px gray;
}

.inset{
  box-shadow:inset 0 0 15px gray;
}

/* Product Card */

.product{
  width:250px;
  margin:30px auto;
  padding:20px;
  background:white;
  box-shadow:3px 3px 10px gray,-3px -3px 10px white;
  transition:.3s;
}
```

```
.product:hover{
  transform:translateY(-5px);
  box-shadow:5px 5px 20px gray;
}

/* Dashboard */

.cards{
  display:flex;
  justify-content:center;
  gap:20px;
  flex-wrap:wrap;
}

.dash-card{
  width:180px;
  height:150px;
  background:white;
  line-height:150px;
  box-shadow:0 0 10px gray;
  transition:.3s;
}

.dash-card:hover{
  transform:translateY(-5px);
  box-shadow:0 0 20px black;
}
```

